Biodegradable Chitosan–Silver Nanocomposite Films as Antibacterial Wound Dressings

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Abstract:

Chronic wounds, including diabetic ulcers and trauma-related injuries, remain difficult to manage due to recurring bacterial infections that significantly delay the healing process and increase the risk of severe complications. Conventional dressings often act only as protective barriers and lack active antimicrobial or regenerative functionality, creating the need for multifunctional biomaterials. In this work, a biodegradable chitosan–silver nanocomposite film was developed as an advanced wound dressing with both antibacterial and tissue-healing properties. Chitosan, a natural polysaccharide, provides excellent biocompatibility, biodegradability, and hemostatic effects, while silver nanoparticles impart strong broad-spectrum antimicrobial activity against common wound pathogens. The films were fabricated using a solvent-free casting process, ensuring uniform nanoparticle distribution and minimizing environmental impact. Material characterization confirmed that the films possessed improved tensile strength and flexibility compared to pure chitosan films, while antibacterial assays demonstrated significant inhibition against Escherichia coli and Staphylococcus aureus. Furthermore, cytocompatibility studies using fibroblast cells indicated high cell viability and adhesion, validating the material's safety for biomedical use. By combining mechanical stability, infection control, and biodegradability, this nanocomposite shows promise as a scalable, cost-effective, and eco-friendly alternative to conventional wound dressings. Its dual functionality makes it especially relevant not only for public healthcare systems but also for defense medicine and emergency trauma care, where rapid wound healing and infection prevention are vital.

Keywords:

Biomaterials; Chitosan; Silver Nanoparticles; Nanocomposites; Wound Healing